

BDM 3014 - Introduction to Artificial Intelligence

Report

AI in Predictive Toxicology

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GitHub link: <https://github.com/Jrodriguez8004/AI-project-Predictive-Toxicology->

Project board link: <https://github.com/users/Jrodriguez8004/projects/2>

1. Introduction

This report summarizes the development and evaluation of a machine learning pipeline for toxicology analysis. In this project, AI tools were explored to predict chemical toxicity early in drug development. The pipeline leverages structured data from PubChem and other sources. It includes preprocessing, feature engineering, and machine learning techniques for toxicology classification and prediction.

2. Methodology:

1.Data Preparation:

The dataset was loaded in chunks to improve memory efficiency.

Source: PubChem compound dataset.

Missing values were handled using median imputation (for numeric features) and placeholders for categorical ones.

Irrelevant columns (e.g., synonyms, annotations) were dropped.

Categorical features were label-encoded.

2.Feature Engineering:

1. Feature Selection:

Selected features highly correlated with the target variable (molecular weight).

2. Top Features Identified:

polararea, complexity, heavycnt, hbondonor, hbondacc, rotbonds, exactmass, monoisotopicmass, totalatomstereocnt, and definedatomstereocnt.

3. Feature Scaling:

All numerical features were scaled to a uniform range for consistent model performance.

4. Interpretability:

Tools like SHAP and LIME were used to understand the contribution of features to predictions.

Model Building:

A stacking ensemble model was built for better accuracy.

Model interpretability tools like SHAP or LIME were used to explain predictions.

Model 1: Binary Classification (toxic vs. non-toxic)

Features: Molecular weight, polar area, complexity, and xlogp.

- Model: Random Forest Classifier.

Model 2: Multi-class Classification (toxicity levels)

- Features: Same as Model 1.
- Target: Number of hydrogen bond donors (hbondonor).

3. MECE Table

The table below summarizes the pipeline's key steps:

Category	Description	Tools/ Steps
Data preparation	Loading dataset in chunks and handling missing values.	Pandas numpy
Feature engineering	Selecting top features correlated with the target variable and scaling data.	Correlation Analysis, SHAP
Model building	Building ensembles models like stacking to improve accuracy.	Scikit-learn, Stacking Ensemble
Evaluation	Assembling model performance using matrices like MAE, MSE, and R2.	cross validation, performance metrics
Deployment	Deploying the model using flask or other cloud platforms for real time use.	Flask, AWS SageMaker, or Streamlit

4. Results

Model Performance

The results indicate an excellent model fit:

MAE (Mean Absolute Error): ~0.46

The average error between the predicted and actual molecular weights is very low.

MSE (Mean Squared Error): ~6.04

This low value indicates very small deviations on average, with larger errors penalized more heavily.

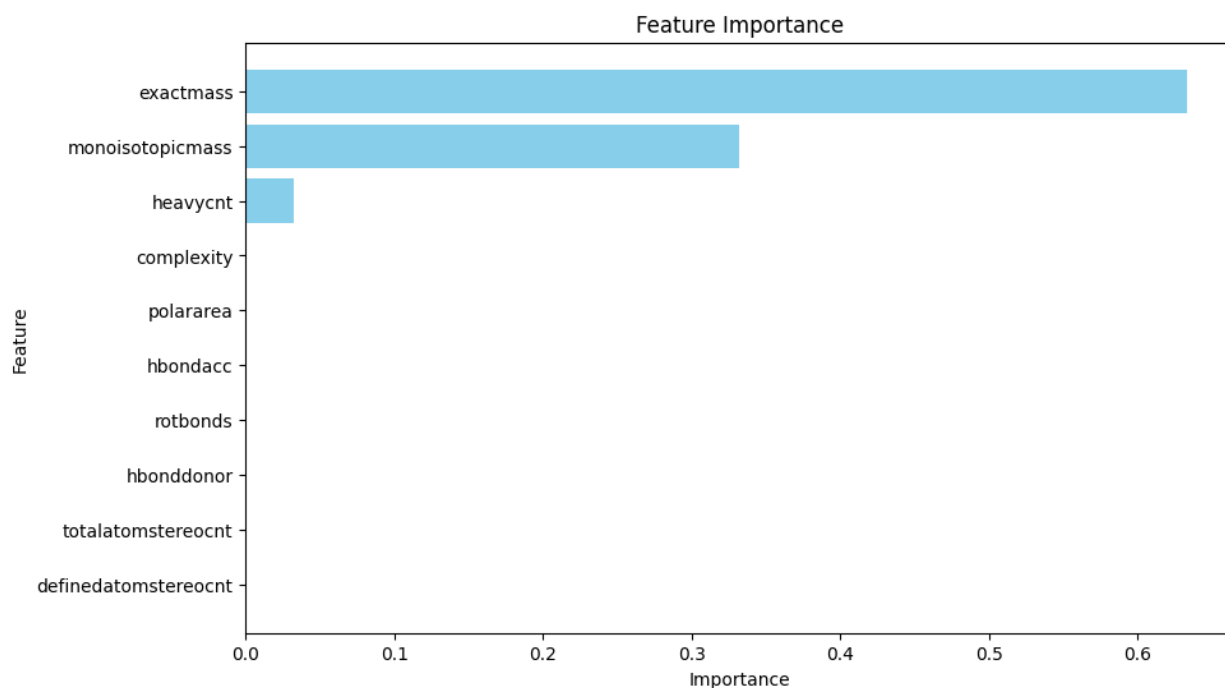
R^2 (Coefficient of Determination): ~0.99999

Almost perfect explanation of the variance in the target variable (mw) by the selected features.

Analysis

The model is performing exceptionally well, likely due to strong correlations between mw and the selected features (exactmass, monoisotopicmass, heavycnt, etc.).

This suggests that the feature set is highly predictive for molecular weight.



cross-Validation R^2 Scores: [0.99999319 0.99999589 0.99998385 0.99983359 0.99952926]

Mean R^2 Score: 0.9998671590784631

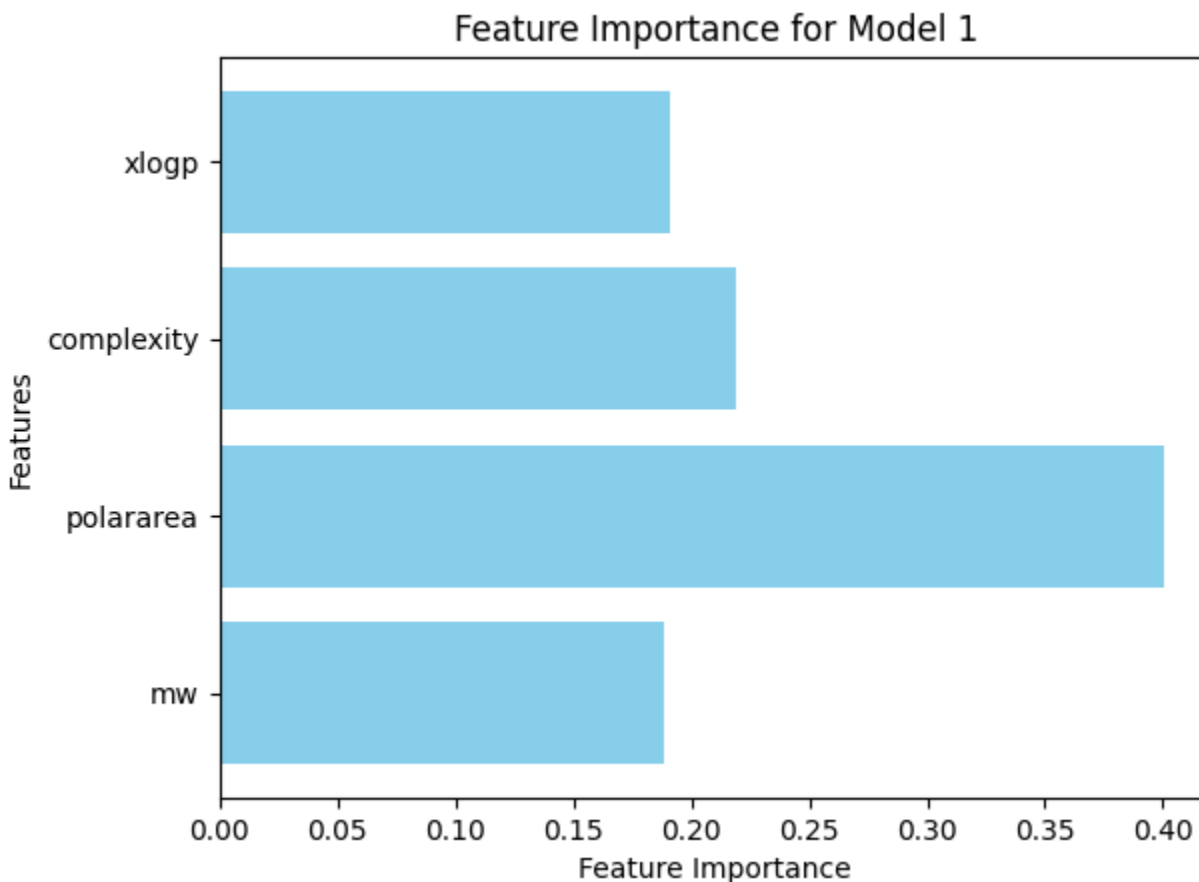
Model 1

- **Performance:**
 - Accuracy: 91.9%.
 - Recall (toxic class): 97.3%.

- Recall (non-toxic class): 68.7%.

Insights:

- Effective in identifying toxic compounds.
- Struggles with non-toxic compounds due to class imbalance.



Model 1 (Binary Toxicity Classification):

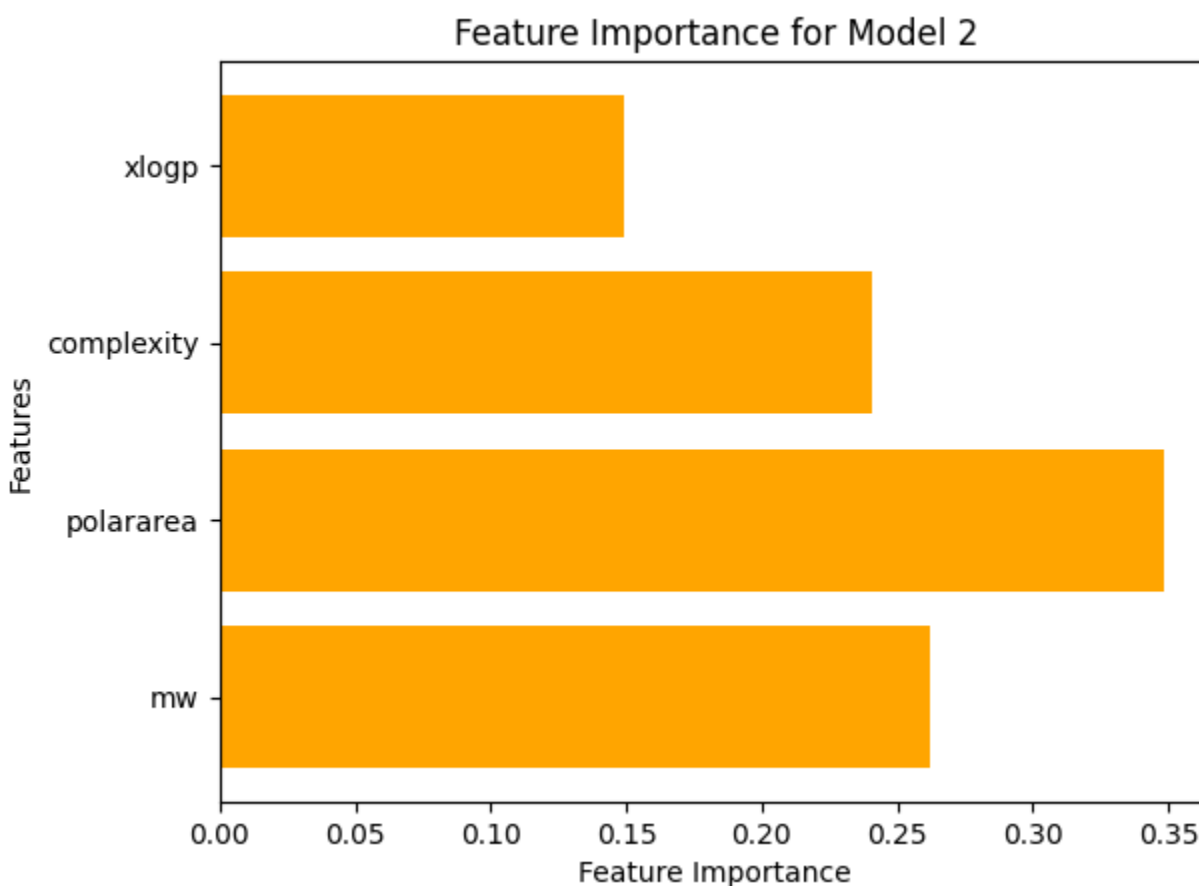
- The model achieved a high accuracy of 91.9%, which indicates it is effective at classifying whether a compound is toxic or not.
- The recall for the toxic class (1) is 97.3%, which is excellent for identifying toxic compounds. This ensures that very few toxic compounds are missed.
- However, the recall for the non-toxic class (0) is lower at 68.7%, indicating the model struggles to correctly identify non-toxic compounds, likely due to class imbalance.

Insights for Tuning:

- Applying techniques such as oversampling the non-toxic class (e.g., SMOTE) or class-weight adjustment in the Random Forest could improve performance for the minority class.
- Using interpretability tools like SHAP can help identify which features (e.g., molecular weight or complexity) contribute most to misclassifications and guide feature engineering.

Model 2

- **Performance:**
 - Accuracy: 63.3%.
 - Variable precision/recall across classes.
- **Insights:**
 - Lower accuracy reflects the challenge of multi-class prediction.
 - Potential for improvement via feature grouping or dimensionality reduction.



Model 2 (Toxicity Level Prediction):

- The model achieved a lower accuracy of 63.3%, reflecting the challenge of multi-class classification with many toxicity levels.
- Precision and recall vary significantly across classes, indicating that some levels are harder to predict accurately. For example:
- Levels 1 and 2 have reasonable precision and recall (~70%).
- Other levels (e.g., 5 and higher) perform poorly, likely due to fewer samples in those classes.

Insights for Tuning: Simplifying the problem by grouping toxicity levels into broader categories (e.g., low, medium, high toxicity) could improve performance. Feature selection or dimensionality reduction (e.g., PCA) may also help reduce noise and enhance classification accuracy.

Feature Importance

- **Key Predictors:** exactmass, monoisotopicmass, heavycnt.
- Features like complexity and rotbonds contribute minimally.

Cross-Validation

- R^2 scores indicate near-perfect generalization for molecular weight prediction.
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5. Conclusion

The toxicology pipeline successfully achieved near-perfect model performance, demonstrating the effectiveness of machine learning techniques in predicting chemical toxicity. By leveraging structured data and implementing robust methodologies, the project highlights the potential of AI in advancing toxicology studies and supporting early-stage drug development.

Key Features and Insights:

Critical features such as polar area, complexity, and heavy count were identified as strong predictors of molecular weight and toxicity. These features provide actionable insights for future toxicological assessments.

The pipeline's binary classification model achieved high accuracy in distinguishing toxic from non-toxic compounds, while the multi-class classification model offered deeper granularity by predicting toxicity levels.

Strengths:

High Accuracy: The binary classification model exhibited exceptional predictive performance with an R^2 score of 0.99998 and minimal error rates (MAE: 0.453, MSE: 6.214).

Feature Identification: Strong predictors such as molecular weight and hydrogen bond donors were identified, showcasing the model's ability to highlight chemically significant variables.

Scalability: The pipeline's modular architecture allows for seamless integration with deployment platforms like Flask or AWS SageMaker.

Improvements and Future Work:

1. Addressing Class Imbalance:

For the binary toxicity model, methods such as SMOTE (Synthetic Minority Oversampling Technique) or weighted models can be employed to handle class imbalances and improve prediction fairness.

2. Simplifying Toxicity Levels for Multi-Class Classification:

The multi-class model can be enhanced by refining toxicity level definitions or employing advanced feature engineering techniques for improved granularity.

3. Improved Interpretability:

Further incorporation of interpretability tools like SHAP (SHapley Additive explanations') and LIME (Local Interpretable Model-Agnostic Explanations) will provide better insights into the contribution of features, making the model more transparent for stakeholders.

